

Plastics

Thermoplastics

Selection list

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Foreword

The German version is binding.

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This Group Standard has been coordinated with the responsible departments of the BMW Group. Author, approver and reviewer are stored in and can be retrieved from the TEREK system master data.

Amendments

The following amendments have been made to GS 93016:2020-03:

- listing of materials from section 4 and 5 deleted and now managed in B2B-Portal;
- section 3.1 and 3.2 added;
- section 3.2 (now 4.2) changed;
- section 3.3 (now 4.3) changed;
- section 3.4 (now 4.4) changed;
- section 6 (now 5) changed;
- cited standards updated;
- Group Standard editorially revised.

Previous editions

BMW(S) 60100.0: 1980-10, 1984-07, 1987-05, 1989-08, 1990-11, 1992-01, 1994-01, 1995-11, 1998-03

GS 93016: 2003-09, 2006-07, 2008-07, 2011-01, 2013-01, 2015-08, 2017-07, 2018-09, 2020-03

1 Scope and purpose

This Group Standard presents a selection of thermoplastic injection molding compounds and lists the names of the raw material producers along with their trade names.

2 Normative references

This Group Standard incorporates provisions from other publications. These normative references are cited at the appropriate places in the text and the publications are listed hereafter. The respective latest edition of the publication is applicable.

DIN EN ISO 1043-1	Plastics; Symbols and abbreviated terms; Part 1: Basic polymers and their special characteristics
DIN EN ISO 14021	Environmental labels and declarations; Self-declared environmental claims (Type II environmental labelling)
GS 91003	Marking of parts; Marking of materials
GS 91021	Virtual validation; Determination of material data for crash simulation; Thermoplastic materials
GS 93008-1	Substances of concern; Materials and components; Prohibited and declarable substances
GS 93055	Materials in purchase parts; Selection, Specification, Test and release

3 Abbreviations and codes

3.1 Abbreviations

B2B	Business-to-business
PVD	Physical vapor deposition
SOP	Start of production
WS	Material specification

3.2 Symbols and abbreviated terms

The symbols and abbreviated terms are based on DIN EN ISO 1043-1.

4 Delivery specifications and instructions

4.1 Delivery specifications

All materials listed in this Group Standard shall fulfill the requirements according to GS 93008-1.

Parts made of polymer materials shall always bear a material identification mark according to GS 91003. The symbol is indicated in this Group Standard next to the name of the plastic (> ... <).

Obligation to indicate changes of the material composition (recipe):

- quantitative change: Change of composition of a product and adding or removing of components;
- qualitative change: Change of the physical, chemical or toxicological parameter and/or characteristics of a product.

The use of plastics that are stabilized with halides (e.g. for heat stabilization) for housings and housing parts of electrical and electronic parts is only permissible in accordance with the following halide-limit values:

- components with open semiconductor max. 10 ppm;
- components with enclosed semiconductors max. 100 ppm.

This is also valid for the superordinate component module or for the assembly of parts.

4.2 Notes

The selection of materials for components for which the supplier signs responsible (purchase parts) is not governed by GS 93016.

For components for which BMW signs responsible, also thermoplastics not listed in GS 93016 but released through other technical regulations (e.g. WS) may be used.

For purchase parts, the selection, specification, testing, documentation and release of materials is governed in GS 93055.

The selection of materials according to GS 93016 does not guarantee the function of parts manufactured from these materials. Meeting the requirements in the award documents is decisive (e.g. requirements specification).

The specialist department „Materials, Process Technology“ shall be involved in the material selection for new parts with BMW responsibility (in-house parts, "built to print", materials as directed parts).

The list ensures the required mechanical properties of the material groups. The material properties and specifications are stored in the BMW Materials database.

The design and production process (e.g. joint lines, wall thicknesses, radii, rib connections), environmental conditions (e.g. temperature spectra, media, UV load) and component-specific requirements (e.g. load spectrum, paintability) shall be ensured in series development.

The release in GS 93016 does not mean that the downstream processes (e.g. paint coat, electroplating, PVD coatings, lamination, overmolding, welding) are released on principle. A component-specific release according to the requirements specification is required in all cases.

Responsibility for defining the shrinkage lies with the part supplier.

The processing specifications of the raw material manufacturers shall be complied with.

Injection-molded BMW texturing initial sample panels are to be used as a reference for the surface texture of the released BMW grains. The manufacturer who produces the texturing finish shall know the steel quality of the injection mold as well as the material for the plastic part.

A selection of thermoplastics from the selection list does not guarantee compliance with emissions requirements for parts manufactured from these materials. The emission values of the part are decisive. In order to meet the emission targets, granulates for interior trim parts shall be pre-treated by annealing, if required.

Preference of use to be given to original dyed materials. Dyeing using master batch shall be harmonized with specialist department „Materials, Process Technology“.

Master batches can have a significant influence on the material properties. The system supplier assumes complete responsibility for the material if additives are added to the basic material (e.g. color master batch). If justified reasons exist for the use of master batches, these shall be matched with the respective polymer basis (no polyolefin master batch for ABS and PA, no universal master batch). Particular care shall be given to prevent that the master batch and other additives in the master batch like UV-stabilizers do not significantly alter the material properties (e.g. embrittlement after alternating climate test). The dosing specifications of the master batch manufacturers and common, permitted fluctuation ranges shall be adhered to and verified by the use of suitable dosing systems (e.g. gravimetric and controlled systems). Each color requires a color release in comparison with the color initial sample plate.

In newly-injected, dry state, parts made of polyamides often lack sufficient tenacity, e.g. assembly of clip connections. In particular in dry autumn and winter months this may result in assembly problems due to the lower moisture absorption of the polyamide parts. Adding water during transport of the parts or immersion of the parts in water before assembly increases the tenacity significantly. An alternative would be the use of an impact-modified (impact resistant in dry condition) polyamide.

Any change in materials after the SOP shall be reported to the releasing specialist department and shall be agreed upon with the Quality Control Centers involved and the specialist department "Materials, Process Technology".

4.3 Notes on adding materials to GS 93016

Preconditions:

- agreement with the "Materials, Process Technology" specialist department;
- demonstrated series application at BMW or existing material card according to GS 91021;
- release test by a laboratory qualified by BMW;
- the test specimens shall be created by the selected test laboratory; the processing parameters shall be documented;
- the use of the laboratory data (material data) by BMW requires approval;
- the raw material supplier shall bear the costs of and commission the tests;
- listing of the material trade name in this Group Standard by the raw material manufacturer is voluntary;
- for each trade name an expiration date is noted. An extension is only granted if a current BMW series application is demonstrated by the raw material supplier.

The approved laboratories are defined via the materials specialist department and may be retrieved in their latest applicable version via the B2B-Portal:

B2B > Departments > Development > Materials > Thermoplastics

4.4 Latest selection list

The selection list is defined by the materials specialist department and may be retrieved in their latest applicable version via the B2B-Portal:

B2B > Departments > Development > Materials > Thermoplastics

The following trade names do not release the suppliers either from compliance with the quality control provisions applicable for these products nor from the indemnity obligation in the event of infringement on an industrial property right of any third party ensuing from contractual use of the products.

Production sites of the approved material trade names upon integration into GS 93016:

AM America
AP Asia Pacific
EC European countries

5 Recycled materials, regenerate materials and regrind

Definition: recycled materials are materials that stem from one of the subsequent recycling processes - utilization of residual production material (post industry and closed loop) or secondary material/utility waste (post consumer) - and serve for the substitution of primary materials.

Recycled materials are distinguished as follows:

- regrind is obtained by grinding plastics. Regrind has different and irregular particle sizes from 2 mm to 5 mm and may contain dust;
- regranulate is obtained from regrind that is subjected to a melting process. This material has a uniform particle size and contains no dust;
- regenerate material is obtained by melting (compounding) and addition of additives to improve the characteristics. Regenerate material has a uniform particle size, no dust content and specific characteristics. The use of regenerate material is subject to a delivery specification between recycled material producer and part manufacturer for all relevant criteria.

In the general context, only the generic term recycled material will be used.

The recycled material content is the mass portion of the recycled material in proportion to the total mass of thermoplastics in a product (see DIN EN ISO 14021).

For series release, parts shall be validated with the maximum planned recycled material content (according to part specific requirements specification and/or engine test stands and vehicles.

POM recycled material shall not be used in the interior due to formaldehyde emissions requirements.